YEAR 9 MATHS · CHAPTER 5 · MIND MAP

Measurement

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1 Errors and accuracy

A measurement is accurate to plus or minus half the smallest unit.

12 cm to the nearest cm: between **11.5 cm** and **12.5 cm**

Greatest possible error: 0.5 cm

2 Length and perimeter

Perimeter is the distance around the outside: add every side.

1 km = 1000 m, 1 m = 100 cm, 1 cm = 10 mm

Rectangle 8 by 3: $P = 2(8 + 3) = 22$

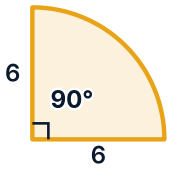
Watch out: convert to the same unit before you add.

3 Circumference and sectors

$C = \pi d = 2\pi r$. A sector is a fraction $\frac{\theta}{360}$ of a circle.

$r = 5$: $C = 10\pi \approx 31.4$

Sector $r = 6$, $\theta = 90°$: $P = 2(6) + \frac{90}{360} \times 2\pi(6) \approx 21.4$



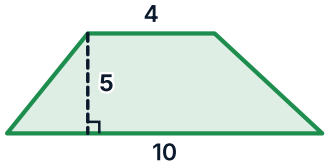
4 Area

Rectangle lw · triangle $\frac{1}{2}bh$ · parallelogram bh · trapezium $\frac{1}{2}(a + b)h$ · circle πr^2

Trapezium $a = 4$, $b = 10$, $h = 5$: $\frac{1}{2} \times 14 \times 5 = 35$

Circle $r = 3$: $9\pi \approx 28.3$

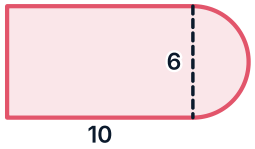
1 m² = 10 000 cm² (100 × 100).



5 Composite shapes

Split into shapes you know and add, or take a shape away.

10 by 6 rectangle with a semicircle on the 6 side: $60 + \frac{1}{2}\pi(3^2) \approx 74.1$



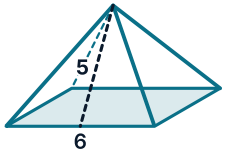
Watch out: the semicircle's radius is half the side: 3, not 6.

6 Surface area of prisms and pyramids

Add the areas of every face. Drawing the net helps.

Cube with side 4: $6 \times 16 = 96$

Square pyramid, base 6, triangle height 5: $36 + 4 \times \frac{1}{2} \times 6 \times 5 = 96$

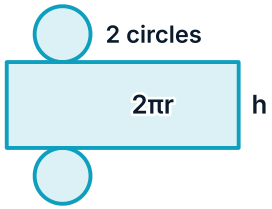


7 Surface area of cylinders

$SA = 2\pi r^2 + 2\pi rh$ (two circles plus the curved side).

$r = 3$, $h = 10$: $18\pi + 60\pi = 78\pi \approx 245.0$

An open cylinder (like a cup) has only one circle.

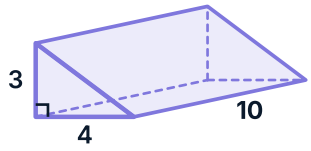


8 Volume of prisms

$V = Ah$: area of the cross-section times the length.

Triangular prism, triangle $\frac{1}{2} \times 4 \times 3 = 6$, length 10: $V = 60$

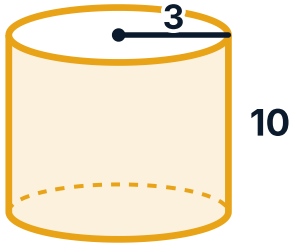
1 cm³ = 1 mL, 1000 cm³ = 1 L, 1 m³ = 1000 L



9 Volume of cylinders

$V = \pi r^2 h$: the circle's area times the height.

$r = 3$ cm, $h = 10$ cm: $V = 90\pi \approx 282.7$ cm³, about 283 mL



Watch out: use the radius, not the diameter.